

D'Alembert's Solution to the 1D Wave Equation

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About

Jean-Baptiste le Rond d'Alembert was a French mathematician, mechanician, physicist, philosopher, and music theorist. Until 1759 he was, together with Denis Diderot, a co-editor of the Encyclopédie. D'Alembert's formula for obtaining solutions to the wave equation is named after him. [Wikipedia](#)

Born: November 16, 1717, Paris, France

Died: October 29, 1783 (age 65 years), Paris, France

Books: Preliminary Discourse to the Encyclopedia of Diderot · See more

Education: University of Paris (1738), Collège des Quatre-Nations (1735), Paris Law Faculty

Parents: Claudine Guérin de Tencin, Louis-Camus Destouches

Awards: ForMemRS (1748)

Notable students: Pierre-Simon Laplace

5. D'Alembert's Solution: Travelling Waveform

3.1 Change of variables: Let $v = x + ct$, $w = x - ct$

$$\text{Then } \frac{\partial v}{\partial x} = 1, \quad \frac{\partial v}{\partial t} = c$$

$$\frac{\partial w}{\partial x} = 1, \quad \frac{\partial w}{\partial t} = -c$$

3.2 Chain Rule: For $u(x, t) = u(x(v, w), t(v, w))$
we have

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial v} \frac{\partial v}{\partial x} + \frac{\partial u}{\partial w} \frac{\partial w}{\partial x} = \frac{\partial u}{\partial v} + \frac{\partial u}{\partial w}$$

Hence:

$$\begin{aligned} \frac{\partial^2 u}{\partial x^2} &= \frac{\partial}{\partial x} \left(\frac{\partial u}{\partial x} \right) = \frac{\partial}{\partial x} \left(\frac{\partial u}{\partial v} + \frac{\partial u}{\partial w} \right) \\ &= \frac{\partial^2 u}{\partial v^2} \frac{\partial v}{\partial x} + \frac{\partial^2 u}{\partial w \partial v} \frac{\partial w}{\partial x} + \frac{\partial^2 u}{\partial v \partial w} \frac{\partial v}{\partial x} + \frac{\partial^2 u}{\partial w^2} \frac{\partial w}{\partial x} \\ &= u_{vv} + u_{vw} + u_{wv} + u_{ww} \end{aligned}$$

Clairut's Theorem: If $u(x(v, w), y(v, w))$ is "nice", then $\frac{\partial^2 u}{\partial v \partial w} = \frac{\partial^2 u}{\partial w \partial v}$

$$\text{Hence: } u_{xx} = u_{vv} + 2u_{vw} + u_{ww}$$

Exercise: Show that $u_{tt} = c^2(u_{vv} - 2u_{vw} + u_{ww})$

3.3 Transforming the wave equation

Starting with $u_{tt} = c^2 u_{xx}$

we get:

$$c^2(u_{vv} - 2u_{vw} + u_{ww}) = c^2(u_{vv} + 2u_{vw} + u_{ww})$$

$$\text{Which simplifies to: } u_{vw} = 0 \Rightarrow \frac{\partial^2 u}{\partial \omega \partial v} = 0$$

3.4 Solving the wave equation:

$$\text{from } \frac{\partial^2 u}{\partial \omega \partial v} = 0 \Rightarrow \int_{v=\text{const}} \frac{\partial}{\partial \omega} \left(\frac{\partial u}{\partial v} \right) d\omega = \int_{v=\text{const}} 0 d\omega$$

$$\Rightarrow \frac{\partial u}{\partial v} = h(v)$$

$$\text{Then } u = \int_{\omega=\text{const}} \left(\frac{\partial u}{\partial v} \right) dv = \int_{\omega=\text{const}} h(v) dv = \underbrace{\Phi(v) + \Psi(\omega)}_{\text{unknown, depends on boundary and initial conditions}}$$

3.5 Using IC:

$$u(x, t) = \Phi(x+ct) + \Psi(x-ct)$$
$$\Rightarrow u_t(x, t) = c\Phi'(x+ct) - c\Psi'(x-ct)$$

$$\text{So } u(x, 0) = f(x) \Rightarrow \Phi(x) + \Psi(x) = f(x)$$
$$u_t(x, 0) = g(x) \Rightarrow c[\Phi'(x) - \Psi'(x)] = g(x)$$

$$\text{So then } \Phi'(x) - \Psi'(x) = \frac{g(x)}{c}$$

$$\Rightarrow \Phi(x) - \Psi(x) = \frac{1}{c} \int_0^x g(s) ds$$

$$\text{We also have: } \Phi(x) + \Psi(x) = f(x)$$

Simultaneous Equations

$$\text{Then } \Phi(x) = \frac{1}{2} f(x) + \frac{1}{2c} \int_0^x g(s) ds$$

$$\text{and } \Psi(x) = \frac{1}{2} f(x) - \frac{1}{2c} \int_0^x g(s) ds$$

$$\text{and } \psi(x) = \frac{1}{2}f(x) - \frac{1}{2c} \int g(s) ds$$

Thus

$$u(x, t) = \Phi(x+ct) + \psi(x-ct)$$

$$= \left[\frac{1}{2} f(x+ct) + \frac{1}{2c} \int_{x-ct}^{x+ct} g(s) ds \right] + \left[\frac{1}{2} f(x-ct) - \frac{1}{2c} \int_{x-ct}^{x-ct} g(s) ds \right]$$

$$\Rightarrow u(x, t) = \frac{1}{2} \left[f(x+ct) + f(x-ct) \right] + \frac{1}{2c} \int_{x-ct}^{x+ct} g(s) ds$$

3.6 Special case: Zero initial velocity

If $g(x)$ is zero, then

$$\text{D'Alembert: } u(x, t) = \frac{1}{2} \left[f(x+ct) + f(x-ct) \right]$$

On the other hand, from Fourier:

$$u(x, t) = \sum_{n=1}^{\infty} (A_n \cos(\lambda_n t) + B_n \sin(\lambda_n t)) \sin\left(\frac{n\pi x}{L}\right)$$

$$\text{where, if } g=0, \text{ then } B_n = \frac{2}{cn\pi} \int_0^L g(x) \sin\left(\frac{n\pi x}{L}\right) dx = 0 \text{ for all } n$$

$$\Rightarrow u(x, t) = \sum_{n=1}^{\infty} A_n \cos(\lambda_n t) \sin\left(\frac{n\pi x}{L}\right)$$

Trig Identity:

$$\cos A \sin B = \frac{1}{2} \left[\sin(B+A) + \sin(B-A) \right]$$

$$\Rightarrow u(x, t) = \sum_{n=1}^{\infty} \frac{1}{2} A_n \left[\sin\left(\frac{n\pi}{L}(x+ct)\right) + \sin\left(\frac{n\pi}{L}(x-ct)\right) \right] \quad \text{Recall: } \lambda_n = \frac{n\pi c}{L}$$

If $u(x, 0) = f(x)$, then we know:

$$\sum_{n=1}^{\infty} A_n \sin\left(\frac{n\pi x}{L}\right) = f(x)$$

$$\text{so we can claim: } \sum_{n=1}^{\infty} \frac{1}{2} A_n \sin\left[\frac{n\pi}{L}(x+ct)\right] = \frac{1}{2} f(x+ct)$$

$$\text{and } \sum_{n=1}^{\infty} \frac{1}{2} A_n \sin\left[\frac{n\pi}{L}(x-ct)\right] = \frac{1}{2} f(x-ct)$$

Which gives us

Which gives us

$$u(x,t) = \frac{1}{2} [f(x+ct) + f(x-ct)], \text{ matching D'Alembert's solution!}$$

Food for thought: What is so good about D'Alembert's solution?